

NIKE - SHORTS & TIKTOK INSIGHTS

Business Report

A business-focused analysis of Nike's 2025 short-form video performance across YouTube Shorts and TikTok.

Project	nike-shorts-tiktok-insights
Author	Sudiksha Singhvi
Analysis Period	2025
Platforms	YouTube Shorts & TikTok
Approach	SQL + Power BI + Python + Machine Learning

Prepared as the executive/business layer of the capstone project.

1. Executive Summary

This project analyzes Nike's 2025 short-form video activity across YouTube Shorts and TikTok to identify patterns that can support better digital marketing decisions. The analysis examines platform performance, regional reach, content categories, creators, hashtags, title characteristics, publishing timing, trend momentum, devices, traffic sources, and seasonality. A machine-learning layer is also used to segment videos into groups with similar performance characteristics.

The central business objective is to move from simply reporting social-media metrics to creating a repeatable decision framework for content planning, creator selection, campaign targeting, publishing strategy, and future testing.

2. Business Problem

Nike generates a large amount of short-form video content, making it difficult to determine which combinations of platform, market, content type, creator, timing, and distribution channel produce the strongest outcomes. This project addresses that challenge by translating performance data into structured business questions and decision areas.

3. Business Objectives

- Compare YouTube Shorts and TikTok performance.
- Identify the strongest countries and markets.
- Evaluate content category performance.
- Identify high-performing creators.
- Assess hashtag reach and engagement.
- Evaluate the association between emoji usage and engagement.
- Identify stronger publishing windows.
- Understand trend momentum and trend duration.
- Compare device and device-brand performance.
- Evaluate traffic-source effectiveness.
- Understand seasonal performance patterns.
- Perform a data-quality validation.

4. Methodology & Analytical Stack

Layer	Role in the project
SQL	Answers 12 structured business questions using aggregation, CTEs, ranking, window functions, and validation.
Power BI	Transforms the analysis into an interactive executive dashboard with KPI and business views.
Python	Supports data preparation and machine-learning analysis.
Machine Learning	Uses K-Means clustering to segment videos based on content and engagement characteristics.

5. Core Business Analyses

Platform Performance

Compares total views and average engagement rate between YouTube Shorts and TikTok. This provides the foundation for platform-level resource and content strategy decisions.

Regional Hotspots

Ranks the top five countries by views for each platform, helping identify markets that may merit deeper localization, campaign investment, or creator partnerships.

Content Category Performance

Evaluates average completion rate and average watch time across categories such as sports, lifestyle, and fashion to identify content areas that sustain attention.

Creator Impact

Ranks creators by average views and associates performance with creator tier, supporting more evidence-based influencer and creator partnership decisions.

Hashtag Performance

Identifies the top hashtags by total views and evaluates median engagement rate, allowing reach to be considered alongside engagement quality.

Emoji Effect

Compares median engagement per 1K views between videos with and without emojis in titles. This is an association analysis, not proof of causation.

Upload Timing

Analyzes performance by day of week and upload hour to identify candidate publishing windows for further testing.

Trend Momentum

Compares engagement velocity and trend duration by trend type to help balance fast-moving trend participation with more durable content.

Device Analysis

Examines completion rates across device types and brands to identify audience-behavior differences.

Traffic Source Performance

Compares video volume, average views, and median completion rate across discovery sources such as recommendation, search, and direct traffic.

Seasonal Insights

Compares views and engagement across event seasons to support calendar planning around relevant sports, holidays, and promotional periods.

Data Quality

Checks for engagement-total mismatches so that downstream business decisions are based on consistent records.

6. Power BI Business Reporting Layer

The Power BI component packages the analytical work into an executive-friendly reporting layer. The dashboard is intended to help a decision-maker move from high-level KPIs to content, creator, market/device, and trend/seasonality analysis.

- **Executive Overview:** Total views, total videos, average engagement rate, and average completion rate.
- **Content Insights:** Category performance, hashtag performance, emoji impact, and duration versus watch-time analysis.
- **Creator Insights:** Top creators, creator-tier comparison, upload timing, and engagement breakdown.
- **Market & Device Analysis:** Country and region performance, device analysis, and traffic-source comparison.
- **Trend & Seasonality:** Engagement velocity, trend type, trend duration, and seasonal patterns.

7. Machine Learning Business Application

The machine-learning layer applies K-Means clustering to group videos according to a combination of content and engagement characteristics. The feature set includes title length, text richness, like rate, comment rate, share rate, and views per day. The business purpose is to discover naturally occurring content segments that can be studied and targeted differently.

Potential business uses of the clusters:

- Identify segments that combine strong reach with strong engagement.
- Compare whether high-performing videos share common content characteristics.
- Develop different creative approaches for different performance segments.
- Use segment membership as a starting point for future experimentation.

8. Strategic Recommendations

Build a platform-specific content strategy

Evaluate YouTube Shorts and TikTok separately and adapt creative, distribution, and testing plans to each platform's observed performance.

Prioritize high-performing markets

Use country-level performance to guide localization, creator partnerships, and campaign prioritization.

Focus on high-retention content

Use completion rate and watch time together when deciding which content categories deserve greater creative attention.

Use performance-based creator selection

Consider creator-level performance alongside audience size and creator tier when planning partnerships.

Optimize hashtags with both reach and engagement

Avoid judging hashtags only by views; evaluate engagement quality at the same time.

Test publishing windows

Use historical day/hour patterns to design controlled experiments rather than treating timing as a fixed rule.

Respond quickly to high-momentum trends

Use engagement velocity as an input to trend participation while considering trend duration.

Plan around seasonal demand

Align relevant content calendars with major events and seasonal periods where audience attention may shift.

9. Expected Business Impact

Business Area	Expected value
Platform Strategy	Better allocation of content effort across YouTube Shorts and TikTok.
Content Strategy	More deliberate focus on formats and categories that sustain attention.
Creator Strategy	Stronger evidence for creator selection and partnership planning.
Campaign Planning	Better targeting by market, timing, trend, and season.
Decision Making	A repeatable analytics framework for future campaign optimization.

10. Limitations

- The current business report describes the analysis framework and decision areas; it does not invent numerical findings where validated result exports were not available.
- Observed relationships should not automatically be treated as causal relationships.
- Creator, hashtag, timing, and platform performance can be influenced by factors outside the measured variables.
- Future decisions should combine this analysis with controlled experiments and additional campaign context.

11. Conclusion

Nike - Shorts & TikTok Insights demonstrates an end-to-end data analytics workflow that connects business questions with SQL analysis, Power BI reporting, Python, and machine learning. The strongest value of the project is its ability to organize short-form video performance into a decision framework covering platforms, markets, content, creators, timing, distribution, devices, seasons, and content segments.

The next step for a marketing team would be to operationalize these findings through recurring performance monitoring and controlled experiments, turning historical insights into an ongoing optimization process.

Author

Sudiksha Singhvi

Data Analytics | SQL | Python | Power BI | Machine Learning

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